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A Story of Hypofractionation and the Table on the Wall

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I was recently informed that I won a contest that I did not actually enter. A Twitter poll was organized through @Radiation Nation inspired by Sue Yom, the new editor of the journal, asking, “What do people consider the gold standard for treatment planning for hypofractionated RT?” Response options were “HyTEC; AAPM TG-101; Timmerman Sheet; and NRG/RTOG protocols.” With just over 100 total votes in 24 hours, “Timmerman Sheet,” which at University of Texas Southwestern we call our “constraint tables,” won with 37.5% of votes. Not to rub it in, but 2 of the other options, “AAPM TG-101” and most “NRG/RTOG protocols,” were taken directly from older versions of our tables. So, in reality we got 78% of votes. Our team developed the first version of these tables around 2002. The tables were freely disseminated at visiting professor trips, at workshops, at training programs, and via e-mail to physicians, residents, physicists, and dosimetrists. I am told that various versions of the tables are thumbtacked for frequent reference to the walls around dosimetry planning stations across the United States and abroad.

The result of this poll might be surprising. After all, the HyTEC initiative in particular was a formal, almost Herculean effort to provide a thorough consensus guideline for treatment planning of hypofractionated treatment. I made my tables while watching my favorite television shows in the evenings after work. So why didn’t HyTEC prevail in the poll? The obvious answer is that it was just very recently published, and centers will likely gradually adopt its recommendations and adapt to its use. In contrast, our tables have been around nearly 20 years, allowing planners to become comfortable with its format and application. Our tables emerged from the earliest years of SABR such that the story of their conception and evolution has been part of the overall story of SABR and the postmodern use of hypofractionation. For this editorial, I would like to tell my version of the 2, interwoven stories.

SABR, my preferred name over the duller, less descriptive, stereotactic body radiation therapy (or SBRT), is now a commonplace treatment such that centers not performing it are arguably out of touch by missing essential modern capabilities.[1-3] Indeed, with the approach of the dreaded Alternative Payment Model from the Centers for Medicaid and Medicare Services, I think conventionally fractionated radiation therapy will quickly become untenable, spelling doom for treatment regimens that for decades stretched out to 6 to 9 weeks. Instead, this author believes that with no reimbursement incentive, nearly all radiation therapy treatments (save pediatrics, germ cell, and lymphoma) will be short course: SABR or nearly SABR.

I’m gloating, but there was a time not so very long ago that SABR was considered radical, even with utter contempt. The radiation therapy giants who developed and advocated for the competing conventionally fractionated radiation therapy, such as Henry Kaplan, Gilbert Fletcher, Isadore Lampe, and Franz Buschke, were influential in getting Medicare to pay more money for more treatments because they believed passionately in the elegant biology of differential repair between tumors and normal tissue. This conventionally fractionated radiation therapy was so biologically forgiving that it could be used even in the frailest patients. Treatment planning for conventional fractionation was relatively easy because nearly any plan could avoid tissue injury, even sloppy, nonconformal fields that treated huge volumes of normal tissues (eg, anterior posterior/posterior anterior,

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wedge pair). The normal tissue constraints that guided this therapy were, in turn, very simple as well (eg, the single page Emami constraints[4] that I had pinned to my wall before 2000).

In his retirement editorial, Fletcher firmly warned the field not to give large-dose-per-fraction treatments unless the patient was literally days from death.[5] He no doubt had valid, personal, negative experiences with hypofractionation causing the dreaded late effects. SABR, the ultimate hypofractionated threat to conventional radiation therapy, emerged in the mid-1990s via a variety of independent experiences from across the world. All, however, derived from very dramatic improvements in technology, technology that Kaplan, Fletcher, Lampe, and Bushke could hardly have conceived. Rather than exploit the biological elegance of differential repair, SABR simply used technology to compartmentalize tumor from normal tissue. These impressive technologies could have been used to deliver conventionally fractionated treatments (recall protons), but thankfully SABR is not only hypofractionated but, as many emotionally proclaim, extremely hypofractionated.

Although most early experiences with SABR treated metastatic patients at end of life, by 1998 we took the bold step to initiate treatments in patients with early lung cancer who would likely live long enough to actually experience late effects. Our pilot experience showed positive results and was appreciated by open-minded thought leaders.[6] In 2000, our group pitched a trial basically reproducing the earlier Indiana University trials[7] but accruing patients in the large, multicenter consortium, Radiation Therapy Oncology Group (RTOG, now the NRG). Although it took many years to actually activate this first cooperative group trial testing SABR, RTOG 0236,[8,9] the RTOG disease committees wisely negotiated terms for buy-in at enrolling sites. For example, they wanted a list of dose constraints for normal tissues for a 3-fraction course of therapy and some guidelines about how to construct dose and evaluate plans. To most of the committee members, giving 20 Gy × 3 fractions was terrifying. They wanted constraints on paper that, if respected, would provide some confidence and protection. Although we had treated several hundred patients by this time, no such constraints existed. We simply created compact dose distributions with isotropic falloff. Given these firm terms required for activating the trial, and not to be dissuaded, I set out to develop the first ever table of dose constraints for SABR.

Now as a bona fide central nervous system physician, I already had some experience with short-course dose constraints. For radiosurgery, I was very familiar with the often-used Kjellberg isoeffect curve, conveying safe prescription dose for a specific size target (eg, 20 Gy for 2 cm lesion gives only 1% risk of necrosis).[10] Think about this: To report a 1% risk of injury specifically for a 2 cm lesion would require treating many patients, all with 2 cm lesions, to a variable dose (eg, 17-18 Gy, 19-20 Gy, 21-22 Gy) to home in on the reported safe dose of 20 Gy. Imagine how disappointing it was to take a deep dive into the hard-to-find sources leading to the Kjellberg isoeffect curve and learn that not many humans had contributed (mostly mice, monkeys, etc) and considerably fewer than 50 data points generated the entire curve. In fact, the basis for the honored Kjellberg curve was... flimsy. At the time, we all used the curve to prescribe dose for both tumors and arteriovenous malformations. And, despite the reality of its basis, it mostly worked to help radiosurgery professionals avoid toxicity.

Table 1 RTOG 0236 original dose constraints

Organ	Volume	Dose (Gy)
Spinal cord	Any point	18 Gy (6 Gy per fraction)
Esophagus	Any point	27 Gy (9 Gy per fraction)
Ipsilateral brachial plexus	Any point	24 Gy (8 Gy per fraction)
Heart	Any point	30 Gy (10 Gy per fraction)
Trachea and ipsilateral Bronchus	Any point	30 Gy (10 Gy per fraction)

Abbreviation: RTOG = Radiation Therapy Oncology Group.

Back to SABR. A maximum point dose constraint table for 3 fractions was rapidly developed for spinal cord, esophagus, brachial plexus, heart, and trachea, shown in Table 1.[11] In all honesty, except for the spinal cord, all other limits for this original table were my educated guesses. At the time, I asked famous radiobiologist and co-committee member, Jack Fowler, to look over the table, and he kindly replied the constraints were reasonable and didn't ask too many questions. Looking back, the only constraint from that original protocol that has not since significantly changed was our heart maximum point dose of 30 Gy. All of the others were originally too strict (conservative) such that our current constraints, as well as HyTEC, allow higher doses without toxicity. I'm not proud of engineering (sounds better than fabricating) the constraints in a national protocol that ultimately changed the standard of care for treating lung cancer. In my defense, it got things moving in a field that was otherwise pretty stagnant. SABR not only gave a good option to medically inoperable patients with early lung cancer, it also pushed radiation oncology into new frontiers such as treating radioresistant tumors (that's what we used to call hepatoma, melanoma, and kidney cancer), treating metastases of all sorts, and shaming the field into abandoning ridiculously long treatment courses (eg, 9 weeks for prostate cancer). SABR started a hypofractionation revolution of sorts and provided an excitement that the field had a future. None of this could be possible without buy-in, and buy-in required normal tissue constraints.

Of course, the tolerance of normal tissues varies, nonlinearly, with number of fractions. So, we proceeded to make a similar table for 1-, 4-, and 5-fraction SABR and added partial volume limits. The field was very interested in modeling treatment effects at the time, and we decided to try our hand at it. Many argued in favor of using the popular linear-quadratic (LQ) model[12] to simply convert our 3-fraction table into the other SABR options. However, that exercise quickly showed problems as the LQ model clearly overpredicted the potency of 1-fraction treatments in particular. This should have been no surprise as the LQ model was developed for conventional fractionation (ie, small doses within the shoulder of the survival curve). So, we compromised by developing a "universal" model combining the LQ with the much older multitarget model.[13] The latter had plenty of useful biology data from the 1960s and 1970s describing the nature of cell kill past the shoulder,[14] and we all know the alpha/beta ratios pertinent to the LQ within the shoulder. With the universal model, we developed constraints for 1, 3, 4, 5, and 8 fractions. We even had more mature data this time for mouse and pig spinal cord tolerance from van der Kogel and Medin and for brachial plexus from Forquer. This allowed some legitimate interpolations and better extrapolations.

We foolishly decided to publish the universal model in the *International Journal of Radiation Oncology • Biology • Physics*,[15] which led to a second furor in the radiation therapy community. What we thought was an innocent application of established mathematical models caused angst and even outrage. Already abandoning conventionally fractionated radiation, many believed we were now abandoning the time-tested LQ.[16-19] We feebly defended the universal model while strongly arguing that the key to better SABR safety was to perform careful dose escalation clinical trials in humans. These toxicity studies were thankfully performed, including within the RTOG,[20] which constituted a much better strategy compared with any mathematical model to define tolerance. As data emerged, we incorporated the results into our tables, admittedly sometimes based on only case reports.

A tricky question to ask about this story is how you can publish a table of constraints that are mostly engineered/modelled. Won't the journal ask for validation or references? Here comes a bit of devious brilliance. Joel Tepper asked me to be the guest editor of a *Seminars in Radiation Oncology* issue on the topic of hypofractionation. I did as asked and assembled a group of thought leaders who provided excellent contributions on the topic. For each *Seminars* issue, the guest editor is expected to write an introduction, which is actually referenced in Index Medicus. I decided to sneak the tables into that introduction, knowing they could never be published any other way. An extended table of constraints found in the article was titled "Mostly Unvalidated Normal Tissue Constraints for SBRT." Now published and referenced,[21] I have easily republished updates of the constraint tables many times by referencing the *Seminars'* introduction.[22-25] Who says cheaters never win?

The tables appropriately separate serial structures (tubes and wires) from parallel structures (organ parenchyma) as shown in Tables 2-11. This distinction is relevant because these tissues have very different strategies for avoiding dysfunction: Serial structures repair aggressively, and parallel structures take advantage of typically huge redundancy and reserve. Although the serial constraints are straightforward (effectively, "don't cross this line"), the parallel constraints follow the critical volume formalism and are best explained by Schefter et al.[26] and Ritter et al.[27] Rightly so, no mean dose limits appear in the tables. For highly conformal, gross-tumor-only treatments often just barely contacting adjacent structures, mean dose limits are easily shown to be meaningless for affecting toxicity.

I'm personally proud that our constraint tables have contributed to the re-emergence of hypofractionation, and especially SABR. But a word of warning: Constraints, as the name implies, are confining and restraining, particularly if they take on medical/legal connotations. The climate in 1995 before SABR was bound by too many constraints. This time, I mean constraints discouraging new ideas, new investigations, and pursuit of new opportunities. The fear of hypofractionation drew from a clearly wrong perception about dose and late normal tissue injury. Flashing forward to 2021, the maximum point dose constraints in particular from both my tables and Hy-TEC are undoubtedly wrong. They are useful only in relation to technology and approaches as exist today. With the future development of technologies and methods capable of creating, for example, ultrathin pencil beams, I predict we will be able to at least double the maximum dose tolerated by the spinal cord. Doing so would allow opportunity to eradicate tumors that today are rarely controlled. But it won't happen if we constrain such innovation, putting handcuffs on investigators who would otherwise explore better treatments. That's the danger of treating constraints like a religion derived from our current training and experience.

I'm grateful to the many physicians and physicists who have sent me toxicity reports that shaped the tables over the years. Of course, I'm sure the HyTEC and its revisions will eventually supplant our tables, and rightly so. But for now, we're relishing the victory in the virtual contest we never entered. #TimmermanTables

Table 2 Timmerman tables, 1 fraction—Timmerman, 8-2021

Serial tissue	Volume, cm3	Volume max (Gy)	Max point dose (Gy)[*]	Endpoint (grade ≥3)
Optic pathway	<0.2	8	10	Neuritis
Cochlea			9	Hearing loss
Brain stem (not medulla)	<0.5	10	15	Cranial neuropathy
Spinal cord and medulla	<0.35	10	14	Myelitis
Cauda equina	<5	14	16	Neuritis
Sacral plexus	<5	14.4	16	Neuropathy
Esophagus[†]	<5	20	24	Esophagitis
Brachial plexus	<3	13.6	16.4	Neuropathy
Peripheral (named) nerve	<2 cm length	16	20	Neuropathy
Heart/pericardium	<15	16	22	Pericarditis
Great vessels	<10	31	37	Aneurysm
Trachea and large bronchus[†]	<4	27.5	30	Impairment of pulmonary toilet
Bronchus, smaller airways	<0.5	17.4	20.2	Stenosis with atelectasis
Rib	<5	28	33	Pain or fracture
Skin	<10	25.5	27.5	Ulceration
Stomach	<5	17.4	22	Ulceration/fistula
Bile duct			30	Stenosis
Duodenum[†]	<5	17.4	22	Ulceration
Jejunum/ileum[†]	<30	17.6	20	Enteritis/obstruction
Colon[†]	<20	20.5	31	Colitis/fistula
Rectum[†]	<3.5	30	33.7	Proctitis/fistula
	<20	23		
Ureter			35	Stenosis
Bladder wall	<15	12	25	Cystitis/fistula
Penile bulb	<3	16		Erectile dysfunction
Femoral heads	<10	15		Necrosis
Renal hilum/vascular trunk	15	14		Malignant hypertension
Parallel tissue	**Critical volume (cm^3)**	**Critical volume dose max (Gy)**	**Max point dose (Gy)[*]**	**Endpoint (grade ≥3)**
Lung (right and left)	1500 for males and 950 for females[‡]	7.2		Basic lung function
Lung (right and left)			V-8 Gy <37%	Radiation pneumonitis
Liver	700[‡]	11.6		Basic liver function
Renal cortex (right and left)	200[‡]	9.5		Basic renal function

[*] "Point" defined as $\leq$0.035 cm^3.
[†] Avoid circumferential irradiation.
[‡] One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Table 3 Timmerman tables, 2 fractions—Timmerman, 8-2021

Serial tissue	Volume	Volume max (Gy)	Max point dose (Gy)*	Endpoint (grade ≥3)
Optic pathway	<0.2 cm^3	11.7	13.7	Neuritis
Cochlea			11.7	Hearing loss
Brain stem (not medulla)	<0.5 cm^3	13	19.1	Cranial neuropathy
Spinal cord and medulla	<0.35 cm^3	13	18.3	Myelitis
Cauda equina	<5 cm^3	18	20.8	Neuritis
Sacral plexus	<5 cm^3	18.5	20.8	Neuropathy
Esophagus†	<5 cm^3	24.3	28.3	Esophagitis
Brachial plexus	<3 cm^3	17.8	21.2	Neuropathy
Peripheral (named) nerve	<2 cm length	21	25.5	Neuropathy
Heart/pericardium	<15 cm^3	20	26	Pericarditis
Great vessels	<10 cm^3	35	41	Aneurysm
Trachea and large bronchus†	<4 cm^3	34.5	38	Impairment of pulmonary toilet
Bronchus, smaller airways	<0.5 cm^3	21.6	25.1	Stenosis with atelectasis
Rib	<5 cm^3	34	41.5	Pain or fracture
Skin	<10 cm^3	28.3	30.3	Ulceration
Stomach	<5 cm^3	20	26	Ulceration/fistula
Bile duct			33	Stenosis
Duodenum†	<5 cm^3	20	26	Ulceration
Jejunum/ileum†	<30 cm^3	19.2	24	Enteritis/obstruction
Colon†	<20 cm^3	25.8	39	Colitis/fistula
Rectum†	<3.5 cm^3	38	41.3	Proctitis/fistula
	<20 cm^3	26.7		
Ureter			37.5	Stenosis
Bladder wall	<15 cm^3	14.5	29	Cystitis/fistula
Penile bulb	<3 cm^3	20.5		Erectile dysfunction
Femoral heads	<10 cm^3	19.5		Necrosis
Renal hilum/vascular trunk	15 cm^3	16.8		Malignant hypertension
Parallel tissue	**Critical volume (cm^3)**	**Critical volume dose max (Gy)**	**Max point dose (Gy)***	**Endpoint (grade ≥3)**
Lung (right and left)	1500 for males and 950 for females‡	9.4		Basic lung function
Lung (right and left)			V-10 Gy <37%	Radiation Pneumonitis
Liver	700‡	15.1		Basic liver function
Renal cortex (right and left)	200‡	12.5		Basic renal function

* "Point" defined as ≤0.035 cm^3.
† Avoid circumferential irradiation.
‡ One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Table 4 Timmerman tables, 3 fractions—Timmerman, 8-2021

Serial tissue	Volume	Volume max (Gy)	Max point dose (Gy)[*]	Endpoint (grade ≥3)
Optic pathway	<0.2 cm^3	15.3	17.4	Neuritis
Cochlea			14.4	Hearing loss
Brain stem (not medulla)	<0.5 cm^3	15.9	23.1	Cranial neuropathy
Spinal cord and medulla	<0.35 cm^3	15.9	22.5	Myelitis
Cauda equina	<5 cm^3	21.9	25.5	Neuritis
Sacral plexus	<5 cm^3	22.5	25.5	Neuropathy
Esophagus[†]	<5 cm^3	27.9	32.4	Esophagitis
Brachial plexus	<3 cm^3	22	26	Neuropathy
Peripheral (named) nerve	<2 cm length	25.5	30.6	Neuropathy
Heart/pericardium	<15 cm^3	24	30	Pericarditis
Great vessels	<10 cm^3	39	45	Aneurysm
Trachea and large bronchus[†]	<5 cm^3	39	43	Impairment of pulmonary toilet
Bronchus, smaller airways	<0.5 cm^3	25.8	30	Stenosis with atelectasis
Rib	<5 cm^3	40	50	Pain or fracture
Skin	<10 cm^3	31	33	Ulceration
Stomach	<5 cm^3	22.5	30	Ulceration/fistula
Bile duct			36	Stenosis
Duodenum[†]	<5 cm^3	22.5	30	Ulceration
Jejunum/ileum[†]	<30 cm^3	20.7	28.5	Enteritis/obstruction
Colon[†]	<20 cm^3	28.8	45	Colitis/fistula
Rectum[†]	<3.5 cm^3	43	47	Proctitis/fistula
	<20 cm^3	30.3		
Ureter			40	Stenosis
Bladder wall	<15 cm^3	17	33	Cystitis/fistula
Penile bulb	<3 cm^3	25		Erectile dysfunction
Femoral heads	<10 cm^3	24		Necrosis
Renal hilum/vascular trunk	15 cm^3	19.5		Malignant hypertension
Parallel tissue	**Critical volume (cm^3)**	**Critical volume dose max (Gy)**		**Endpoint (grade ≥3)**
Lung (right and left)	1500 for males and 950 for females[‡]	10.8		Basic lung function
Lung (right and left)			V-11.4 Gy <37%	Pneumonitis
Liver	700[‡]	17.7		Basic liver function
Renal cortex (right and left)	200[‡]	14.7		Basic renal function

[*] "Point" defined as ≤0.035 cm^3.
[†] Avoid circumferential irradiation.
[‡] One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Table 5 Timmerman tables, 4 fractions—Timmerman, 8-2021

Serial tissue	Volume	Volume max (Gy)	Max point dose (Gy)[*]	Endpoint (grade ≥3)
Optic pathway	<0.2 cm^3	19.2	21.2	Neuritis
Cochlea			18	Hearing loss
Brain stem (not medulla)	<0.5 cm^3	20.8	27.2	Cranial neuropathy
Spinal cord and medulla	<0.35 cm^3	18	25.6	Myelitis
Cauda equina	<5 cm^3	26	28.8	Neuritis
Sacral plexus	<5 cm^3	26	28.8	Neuropathy
Esophagus[†]	<5 cm^3	30.4	35.6	Esophagitis
Brachial plexus	<3 cm^3	24.8	29.6	Neuropathy
Peripheral (named) nerve	<2cm length	28.8	34.8	Neuropathy
Heart/pericardium	<15 cm^3	28	34	Pericarditis
Great vessels	<10 cm^3	43	49	Aneurysm
Trachea and large bronchus[†]	<5 cm^3	42.4	47	Impairment of pulmonary toilet
Bronchus, smaller airways	<0.5 cm^3	28.8	34.8	Stenosis with atelectasis
Rib	<5 cm^3	43	54	Pain or fracture
Skin	<10 cm^3	33.6	36	Ulceration
Stomach	<5 cm^3	25	33.2	Ulceration/fistula
Bile duct			38.4	Stenosis
Duodenum[†]	<5 cm^3	25	33.2	Ulceration
Jejunum/ileum[†]	<30 cm^3	22.4	31.6	Enteritis/obstruction
Colon[†]	<20 cm^3	30.8	48.5	Colitis/fistula
Rectum[†]	<3.5 cm^3	47.2	51.6	Proctitis/fistula
	<20 cm^3	34		
Ureter			43	Stenosis
Bladder wall	<15 cm^3	18.5	35.6	Cystitis/fistula
Penile bulb	<3 cm^3	27		Erectile dysfunction
Femoral heads	<10 cm^3	27		Necrosis
Renal hilum/vascular trunk	15 cm^3	21.5		Malignant hypertension
Parallel tissue	**Critical volume (cm^3)**	**Critical volume dose max (Gy)**		**Endpoint (grade ≥3)**
Lung (right and left)	1500 for males and 950 for females[‡]	12		Basic lung function
Lung (right and left)			V-12.8 Gy <37%	Pneumonitis
Liver	700[‡]	19.6		Basic liver function
Renal cortex (right and left)	200[‡]	16		Basic renal function

[*] "Point" defined as ≤0.035 cm^3.
[†] Avoid circumferential irradiation.
[‡] One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Table 6 Timmerman tables, 5 fractions—Timmerman, 8-2021

Serial tissue	Volume	Volume max (Gy)	Max point dose (Gy)*	Endpoint (grade ≥3)
Optic pathway	<0.2 cm^3	23	25	Neuritis
Cochlea			22	Hearing loss
Brain stem (not medulla)	<0.5 cm^3	23	31	Cranial neuropathy
Spinal cord and medulla	<0.35 cm^3	22	28	Myelitis
Cauda equina	<5 cm^3	30	31.5	Neuritis
Sacral plexus	<5 cm^3	30	32	Neuropathy
Esophagus†	<5 cm^3	32.5	38	Esophagitis
Brachial plexus	<3 cm^3	27	32.5	Neuropathy
Peripheral (named) nerve	<2 cm length	31.5	38	Neuropathy
Heart/pericardium	<15 cm^3	32	38	Pericarditis
Great vessels	<10 cm^3	47	53	Aneurysm
Trachea and large bronchus†	<5 cm^3	45	50	Impairment of pulmonary toilet
Bronchus, smaller airways	<0.5 cm^3	32	40	Stenosis with atelectasis
Rib	<5 cm^3	45	57	Pain or fracture
Skin	<10 cm^3	36.5	38.5	Ulceration
Stomach	<5 cm^3	26.5	35	Ulceration/fistula
Bile duct			41	Stenosis
Duodenum†	<5 cm^3	26.5	35	Ulceration
Jejunum/ileum†	<30 cm^3	24	34.5	Enteritis/obstruction
Colon†	<20 cm^3	32.5	52.5	Colitis/fistula
Rectum†	<3.5 cm^3	50	55	Proctitis/fistula
	<20 cm^3	37.5		
Ureter			45	Stenosis
Bladder wall	<15 cm^3	20	38	Cystitis/fistula
Penile bulb	<3 cm^3	30		Erectile dysfunction
Femoral heads	<10 cm^3	30		Necrosis
Renal hilum/vascular trunk	15 cm^3	23		Malignant hypertension
Parallel tissue	**Critical volume (cm^3)**	**Critical volume dose max (Gy)**	**Max point dose (Gy)***	**Endpoint (grade ≥3)**
Lung (right and left)	1500 for males and 950 for females‡	12.5		Basic lung function
Lung (right and left)			V-13.5 Gy <37%	Pneumonitis
Liver	700‡	21.5		Basic liver function
Renal cortex (right and left)	200‡	17.5		Basic renal function

* "Point" defined as ≤0.035 cm^3.
† Avoid circumferential irradiation.
‡ One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Table 7 Timmerman tables, 8 fractions—Timmerman, 8-2021

Serial tissue	Volume	Volume max (Gy)	Max point dose (Gy)*	Endpoint (grade ≥3)
Optic pathway	<0.2 cm^3	27.2	29.6	Neuritis
Cochlea			26.4	Hearing loss
Brain stem (not medulla)	<0.5 cm^3	27.2	37.6	Cranial neuropathy
Spinal cord and medulla	<0.35 cm^3	26.4	33.6	Myelitis
Cauda equina	<5 cm^3	34	38.4	Neuritis
Sacral plexus	<5 cm^3	34	38.4	Neuropathy
Esophagus†	<5 cm^3	36.8	43.2	Esophagitis
Brachial plexus	<3 cm^3	32.8	39.2	Neuropathy
Peripheral (named) nerve	<2 cm length	37	43	Neuropathy
Heart/pericardium	<15 cm^3	34.4	40	Pericarditis
Great vessels	<10 cm^3	55.2	62	Aneurysm
Trachea and large bronchus†	<5 cm^3	50	56	Impairment of pulmonary toilet
Bronchus, smaller airways	<0.5 cm^3	38.4	48.8	Stenosis with atelectasis
Rib	<5 cm^3	50	63	Pain or fracture
Skin	<10 cm^3	43.2	45.6	Ulceration
Stomach	<5 cm^3	31.2	42	Ulceration/fistula
Bile duct			48	Stenosis
Duodenum†	<5 cm^3	31.2	42	Ulceration
Jejunum/ileum†	<30 cm^3	28.8	40	Enteritis/obstruction
Colon†	<20 cm^3	35.2	57.5	Colitis/fistula
Rectum†	<3.5 cm^3	56	61.5	Proctitis/fistula
	<20 cm^3	45		
Ureter			53	Stenosis
Bladder wall	<15 cm^3	22.4	44.8	Cystitis/fistula
Penile bulb	<3 cm^3	35		Erectile dysfunction
Femoral heads	<10 cm^3	35		Necrosis
Renal hilum/vascular trunk	15 cm^3	28		Malignant hypertension
Parallel tissue	**Critical volume (cm^3)**	**Critical volume dose max (Gy)**		**Endpoint (grade ≥3)**
Lung (right and left)	1500 for males and 950 for females‡	14.4		Basic lung function
Lung (right and left)			V-15.2 Gy <37%	Pneumonitis
Liver	700‡	24.8		Basic liver function
Renal cortex (right and left)	200‡	20		Basic renal function

* "Point" defined as ≤0.035 cm^3.
† Avoid circumferential irradiation.
‡ One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Table 8 Timmerman tables, 10 fractions—Timmerman, 8-2021

Serial tissue	Contouring instructions	Volume	Volume max (Gy)	Max point dose (Gy)	Endpoint (grade ≥3)
Optic pathway	One structure both sides from posterior globe, including chiasm, to proximal optic radiations	<0.5 cm^3	30.6	33.1	Neuritis
Eye (retina)	Each side separately, entire globe	Mean dose	<26	30	Retinitis
Lens	Each side separately			7	Cataract
Eyelid, meibomian glands (one side)	Each side separately, upper and lower lid as 1 structure			21.3	Dry eye syndrome
Lacrimal gland (one side)	Each side separately	<1 cm^3	14.1	23.6	Lack of tears
Cochlea	Each side separately, include at least 3 CT slices	<0.5 cm^3	25	27	Hearing loss
Brain stem (not medulla)	Superiorly from incisura, midbrain and pons only, 1 structure	<5 cm^3	32	38	Cranial neuropathy
Spinal cord	Entire bony canal including at least 10 cm superior and inferior to PTV	<5 cm^3	31	36	Myelitis
Salivary gland (one side)	Each parotid gland separately	<7 cm^3 mean dose	14.1 <17.7	21.3	Xerostomia
Larynx	Starting 1 cm above first appearance of true vocal cord, includes entire cord, arytenoid muscles, corniculate and arytenoid cartilages and portions of thyroid cartilage abutting these structures ending at the first appearance of the cricothyroid ligament	<3 cm^3	30	45	Necrosis/edema
TM joint	Each side separately starting at the superior articular surface near the zygoma bone and ending at the notch at the superior part of the ramus of the mandible	<1 cm^3	37.7	41.4	Inflammation
Cauda equina	Based on the bony limits of the spinal canal starting superiorly at the bottom of the spinal cord (typically around L2) and ending at the inferior extent of the thecal sac (typically S3)	<5 cm^3	35	41	Neuritis
Sacral plexus	Outlining the space defined medially by the sacral foramina from S1-S3, including contouring within the sacral foramina, posteriorly along the limits of the true pelvis, laterally to 2-3 cm lateral to the sacral foramina, and anteriorly about 3-5 mm from the posterior limits of the contour	<5 cm^3	35	41	Neuropathy
Esophagus	Include the mucosal, submucosa, and all muscular layers out to the fatty adventitia at least 10 cm superior and inferior to PTV	<5 cm^3	40	48	Esophagitis
Brachial plexus	Each side separately from the spinal nerves exiting the neuroforamina from around C5 to T2 to include only the major trunks of the brachial plexus using the subclavian and axillary vessels as a surrogate for identifying its location extending proximally at the bifurcation of the brachiocephalic trunk into the jugular/subclavian veins (or carotid/subclavian arteries) and following along the route of the subclavian vein to the axillary vein ending after the neurovascular structures cross the second rib	<3 cm^3	37	43	Neuropathy
Heart/pericardium	Contoured along with the pericardial sac; the superior aspect (or base) for purposes of contouring will begin at the level of the inferior aspect of the aortic arch (aorto-pulmonary window) and extend inferiorly to the apex of the heart	<15 cm^3	36.6	42.5	Pericarditis
Great vessels	The wall and lumen of the named vessel at least 10 cm superior and inferior to PTV	<10 cm^3	55.7	62.9	Aneurysm

(*Continued*)

Table 8 (*Continued*)

Serial tissue	Contouring instructions	Volume	Volume max (Gy)	Max point dose (Gy)	Endpoint (grade ≥3)
Trachea and large bronchus	Contour the trachea and cartilage rings starting 10 cm superior to the PTV and extending inferior to the bronchi ending at the first bifurcation of the named lobar bronchus	<5 cm^3	52	59	Impairment of pulmonary toilet
Skin	The outer 0.5 cm of the body surface anywhere within the whole-body contour.	<10 cm^3	46.3	48.9	Ulceration
Stomach	The entire stomach wall and the gastric contents included from the GE junction to the proximal duodenum at the pyloris	<50 cm^3	33.9	45	Ulceration/fistula
Duodenum	The entire duodenal wall and lumen from the pyloris to the duodenojejunal flexure	<5 cm^3	33.9	45	Ulceration
Jejunum/ileum	Any and all loops of small bowel as 1 structure within 10 cm of the PTV in any direction.	<120 cm^3	33.9	41	Enteritis/ obstruction
Renal hilum/vascular trunk	Each side separately, including major calyces, renal pelvis, and proximal renal artery medial to the aorta	15 cm^3	30.7		Malignant hypertension
Colon	One structure, including wall and contents of lumen starting 10 cm superior to PTV and ending 10 cm below PTV	<20 cm^3	47	60	Colitis/fistula
Rectum (including stool)	One structure, including wall of rectum and all contents in lumen; start contouring 10 cm superior to PTV and then inferior to anal sphincter	<10 cm^3 <20 cm^3 <30 cm^3 <40 cm^3	52 49 46 43	65	Proctitis/fistula
Bladder (with urine)	Contour the bladder wall and all urine ending inferiorly at the base of the prostate	<90 cm^3 <125 cm^3	48 45	53	Cystitis/fistula
Bladder (suprapubic wall)	Contour the anterior inferior wall resting above and around the superior aspect of the pubic bone starting at the prostate inferiorly and extending 2-3 cm superiorly from there	<5 cm^3	23	42	Dysuria
Penile bulb	Contour starting superiorly at the inferior aspect of the pelvic diaphragm (urethral sphincter) and extending inferiorly and anteriorly up to 3 cm	<3 cm^3	38	44	Erectile dysfunction
Femoral heads	Contour both right and left separately	<10 cm^3	38	43.5	Necrosis
Parallel tissue		**Critical volume (cm^3)**	**Critical volume dose max (Gy)**	**Other constraints**	**Endpoint (grade ≥3)**
Lung (right and left) minus GTV	Contour right and left lung as 1, structure including all parenchymal lung tissue but excluding the GTV and major airways (trachea and main/lobar bronchi)	1500 for males and 950 for females*	15		Basic lung function
Lung (right and left) minus GTV	Contour right and left lung as 1 structure, including all parenchymal lung tissue but excluding the GTV and major airways (trachea and main/lobar bronchi)			V-16 Gy <37%	Pneumonitis
Liver minus GTV	Contour right and left lobes as 1 structure, including all parenchymal liver tissue but excluding the GTV and major draining ducts, extrahepatic portal vein, and gall bladder	700 cm^3*	27		Basic liver function
Renal cortex (right and left)	Contour right and left kidney as 1 structure, including all parenchymal capsular tissue but excluding the renal hilum/vascular trunk (see above)	200 cm^3*	21		Basic renal function

* One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Abbreviations: CT = computed tomography; GE = gastroesophageal; GTV = gross target volume; PTV = planning target volume; TM = temporomandibular.

Table 9 **Timmerman tables, 15 fractions—Timmerman, 8-2021**

Serial tissue	Contouring instructions	Volume	Volume Max (Gy)	Max point dose (Gy)	Endpoint (grade ≥3)
Optic pathway	One structure both sides from posterior globe, including chiasm, to proximal optic radiations	<0.5 cm^3	39	42	Neuritis
Eye (retina)	Each side separately, entire globe	Mean dose	<33	37.5	Retinitis
Lens	Each side separately			9	Cataract
Eyelid, meibomian glands (one side)	Each side separately, upper and lower lid as one structure			27	Dry eye syndrome
Lacrimal gland (one side)	Each side separately	<1 cm^3	18	30	Lack of tears
Cochlea	Each side separately, include at least 3 CT slices	<0.5 cm^3	30	33	Hearing loss
Brain stem (not medulla)	Superiorly from incisura, midbrain and pons only, 1 structure	<5 cm^3	40	44	Cranial neuropathy
Spinal cord	Entire bony canal including at least 10 cm superior and inferior to PTV	<5 cm^3	39	42.0	Myelitis
Salivary gland (one side)	Each parotid gland separately	<7 cm^3 Mean dose	18 <22.5	27	Xerostomia
Larynx	Starting 1 cm above first appearance of true vocal cord, include entire cord, arytenoid muscles, corniculate and arytenoid cartilages, and portions of thyroid cartilage abutting these structures ending at the first appearance of the cricothyroid ligament	<3 cm^3	34.5	52.5	Necrosis/edema
TM joint	Each side separately starting at the superior articular surface near the zygoma bone and ending at the notch at the superior part of the ramus of the mandible	<1 cm^3	48	52.5	Inflammation
Cauda equina	Based on the bony limits of the spinal canal starting superiorly at the bottom of the spinal cord (typically around L2) and ending at the inferior extent of the thecal sac (typically S3)	<5 cm^3	40.5	48	Neuritis
Sacral plexus	Outlining the space defined medially by the sacral foramina from S1-S3, including contouring within the sacral foramina, posteriorly along the limits of the true pelvis, laterally to 2-3 cm lateral to the sacral foramina, and anteriorly about 3-5 mm from the posterior limits of the contour	<5 cm^3	40.5	48	Neuropathy
Esophagus	Include the mucosal, submucosa, and all muscular layers out to the fatty adventitia at least 10 cm superior and inferior to PTV	<5 cm^3	45	54	Esophagitis
Brachial plexus	Each side separately from the spinal nerves exiting the neuroforamina from around C5 to T2 to include only the major trunks of the brachial plexus using the subclavian and axillary vessels as a surrogate for identifying its location, extending proximally at the bifurcation of the brachiocephalic trunk into the jugular/subclavian veins (or carotid/subclavian arteries) and following along the route of the subclavian vein to the axillary vein ending after the neurovascular structures cross the second rib	<3 cm^3	48	52.5	Neuropathy
Heart/pericardium	Contoured along with the pericardial sac; the superior aspect (or base) for purposes of contouring will begin at the level of the inferior aspect of the aortic arch (aorto-pulmonary window) and extend inferiorly to the apex of the heart	<15 cm^3	42	48.9	Pericarditis

(*Continued*)

Table 9 (*Continued*)

Serial tissue	Contouring instructions	Volume	Volume Max (Gy)	Max point dose (Gy)	Endpoint (grade ≥3)
Great vessels	The wall and lumen of the named vessel at least 10 cm superior and inferior to PTV	<10 cm^3	57	65	Aneurysm
Trachea and large bronchus	Contour the trachea and cartilage rings starting 10 cm superior to the PTV, extending inferiorly to the bronchi, ending at the first bifurcation of the named lobar bronchus	<5 cm^3	55.5	63	Impairment of pulmonary toilet
Skin	The outer 0.5 cm of the body surface anywhere within the whole body contour	<10 cm^3	54	57	Ulceration
Stomach	The entire stomach wall and the gastric contents included from the GE junction to the proximal duodenum at the pyloris	<50 cm^3	39	51	Ulceration/fistula
Duodenum	The entire duodenal wall and lumen from the pyloris to the duodenojejunal flexure	<5 cm^3	39	51	Ulceration
Jejunum/ileum	Any and all loops of small bowel as one structure within 10 cm of the PTV in any direction	<120 cm^3	39	46.5	Enteritis/ obstruction
Renal hilum/vascular trunk	Each side separately, including major calyces, renal pelvis, and proximal renal artery medially to the aorta	15 cm^3	37.5		Malignant hypertension
Colon	One structure, including wall and contents of lumen starting 10 cm superior to PTV and ending 10 cm below PTV	<20 cm^3	47	64.5	Colitis/fistula
Rectum (including stool)	One structure, including wall of rectum and all contents in lumen; start contouring 10 cm superior to PTV then inferior to anal sphincter	<10 cm^3 <20 cm^3 <30 cm^3 <40 cm^3	60 57 52.5 49.5	70.5	Proctitis/fistula
Bladder (with urine)	Contour the bladder wall and all urine ending inferiorly at the base of the prostate	<90 cm^3 <125 cm^3	55.5 52.5	61.5	Cystitis/fistula
Bladder (suprapubic wall)	Contour the anterior inferior wall resting above and around the superior aspect of the pubic bone starting at the prostate inferiorly and extending 2-3 cm superiorly from there	<5 cm^3	26	48	Dysuria
Penile bulb	Contour starting superiorly at the inferior aspect of the pelvic diaphragm (urethral sphincter) and extending inferiorly and anteriorly up to 3 cm	<3 cm^3	42	48	Erectile dysfunction
Femoral heads	Contour both right and left separately	<10 cm^3	40	46.5	Necrosis
Parallel tissue		**Critical volume (cm^3)**	**Critical volume dose max (Gy)**	**Other constraints**	**Endpoint (grade ≥3)**
Lung (right and left) minus GTV	Contour right and left lung as 1 structure, including all parenchymal lung tissue but excluding the GTV and major airways (trachea and main/lobar bronchi)	1500 for males and 950 for females*	16.5		Basic lung function
Lung (right and left) minus GTV	Contour right and left lung as 1 structure, including all parenchymal lung tissue but excluding the GTV and major airways (trachea and main/lobar bronchi)			V-18 Gy <37%	Pneumonitis
Liver minus GTV	Contour right and left lobes as 1 structure, including all parenchymal liver tissue but excluding the GTV and major draining ducts, extrahepatic portal vein, and gall bladder	700*	30		Basic liver function
Renal cortex (right and left)	Contour right and left kidney as 1 structure, including all parenchymal capsular tissue but excluding the renal hilum/vascular trunk (see above)	200*	24		Basic renal function

* One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Abbreviations: CT = computed tomography; GTV = gross target volume; PTV = planning target volume.

Table 10 Timmerman tables, 20 fractions—Timmerman, 8-2021

Serial tissue	Contouring instructions	Volume	Volume max (Gy)	Max point dose (Gy)	Endpoint (grade ≥3)
Optic pathway	One structure both sides from posterior globe, including chiasm, to proximal optic radiations	<0.5 cm^3	42	48	Neuritis
Eye (retina)	Each side separately, entire globe	Mean dose	<36	42	Retinitis
Lens	Each side separately			10	Cataract
Eyelid, meibomian glands (one side)	Each side separately, upper and lower lid as 1 structure			30	Dry eye syndrome
Lacrimal gland (one side)	Each side separately	<1 cm^3	18	32	Lack of tears
Cochlea	Each side separately, include at least 3 CT slices	<0.5 cm^3	32	36	Hearing loss
Brain stem (not medulla)	Superiorly from incisura, midbrain and pons only, 1 structure	<5 cm^3	44	50	Cranial neuropathy
Spinal cord and medulla	For medulla: starting at inferior pons to foramen magnum; for cord: entire bony canal including at least 10 cm superior and inferior to PTV	<5 cm^3	42	46	Myelitis
Salivary gland (one side)	Each parotid gland separately	<7 cm^3 Mean dose	18 <24	30	Xerostomia
Larynx	Starting 1 cm above first appearance of true vocal cord, include entire cord, arytenoid muscles, corniculate and arytenoid cartilages, and portions of thyroid cartilage abutting these structures, ending at the first appearance of the cricothyroid ligament	<3 cm^3	36	58	Necrosis/edema
TM joint	Each side separately starting at the superior articular surface near the zygoma bone and ending at the notch at the superior part of the ramus of the mandible	<1 cm^3	52	58	Inflammation
Cauda equina	Based on the bony limits of the spinal canal, starting superiorly at the bottom of the spinal cord (typically around L2) and ending at the inferior extent of the thecal sac (typically S3)	<5 cm^3	44	52	Neuritis
Sacral plexus	Outlining the space defined medially by the sacral foramina from S1-S3, including contouring within the sacral foramina, posteriorly along the limits of the true pelvis, laterally to 2-3 cm lateral to the sacral foramina, and anteriorly about 3-5 mm from the posterior limits of the contour	<5 cm^3	44	52	Neuropathy
Esophagus	Include the mucosal, submucosa, and all muscular layers out to the fatty adventitia at least 10 cm superior and inferior to PTV	<5 cm^3	48	58	Esophagitis
Brachial Plexus	Each side separately from the spinal nerves exiting the neuroforamina from around C5 to T2 to include only the major trunks of the brachial plexus using the subclavian and axillary vessels as a surrogate for identifying its location, extending proximally at the bifurcation of the brachiocephalic trunk into the jugular/subclavian veins (or carotid/subclavian arteries) and following along the route of the subclavian vein to the axillary vein ending after the neurovascular structures cross the second rib	<3 cm^3	54	58	Neuropathy
Heart/pericardium	Contoured along with the pericardial sac; the superior aspect (or base) for purposes of contouring will begin at the level of the inferior aspect of the aortic arch (aorto-pulmonary window) and extend inferiorly to the apex of the heart	<15 cm^3	46	52	Pericarditis

(*Continued*)

Table 10 (*Continued*)

Serial tissue	Contouring instructions	Volume	Volume max (Gy)	Max point dose (Gy)	Endpoint (grade ≥3)
Great vessels	The wall and lumen of the named vessel at least 10 cm superior and inferior to PTV	<10 cm^3	60	70	Aneurysm
Trachea and large bronchus	Contour the trachea and cartilage rings starting 10 cm superior to the PTV, extending inferiorly to the bronchi and ending at the first bifurcation of the named lobar bronchus	<5 cm^3	58	66	Impairment of pulmonary toilet
Skin	The outer 0.5 cm of the body surface anywhere within the whole body contour.	<10 cm^3	60	64	Ulceration
Stomach	The entire stomach wall and the gastric contents included from the GE junction to the proximal duodenum at the pyloris	<50 cm^3	42	54	Ulceration/fistula
Duodenum	The entire duodenal wall and lumen from the pyloris to the duodenojejunal flexure	<5 cm^3	42	54	Ulceration
Jejunum/Ileum	Any and all loops of small bowel as 1 structure within 10 cm of the PTV in any direction	<120 cm^3	42	50	Enteritis/ obstruction
Renal hilum/vascular trunk	Each side separately, including major calyces, renal pelvis, and proximal renal artery medially to the aorta	15 cm^3	40		Malignant hypertension
Colon	One structure including wall and contents of lumen starting 10 cm superior to PTV and ending 10 cm below PTV	<20 cm^3	50	66	Colitis/fistula
Rectum (including stool)	One structure including wall of rectum and all contents in lumen; start contouring 10 cm superior to PTV and then inferior to anal sphincter	<10 cm^3 <20 cm^3 <30 cm^3 <40 cm^3	66 62 58 54	74	Proctitis/fistula
Bladder (with urine)	Contour the bladder wall and all urine ending inferiorly at the base of the prostate	<90 cm^3 <125 cm^3	60 56	66	Cystitis/fistula
Bladder (suprapubic wall)	Contour the anterior inferior wall resting above and around the superior aspect of the pubic bone starting at the prostate inferiorly and extending 2-3 cm superiorly from there	<5 cm^3	28	52	Dysuria
Penile bulb	Contour starting superiorly at the inferior aspect of the pelvic diaphragm (urethral sphincter) and extending inferiorly and anteriorly up to 3 cm	<3 cm^3	44	52	Erectile dysfunction
Femoral heads	Contour both right and left separately	<10 cm^3	44	50	necrosis

Parallel tissue		Critical volume (cm^3)	Critical volume dose Max (Gy)	Other constraints	Endpoint (grade ≥3)
Lung (right and left) minus GTV	Contour right and left lung as 1 structure, including all parenchymal lung tissue but excluding the GTV and major airways (trachea and main/lobar bronchi)	1500 for males and 950 for females*	18		Basic lung function
Lung (right and left) minus GTV	Contour right and left lung as 1 structure, including all parenchymal lung tissue but excluding the GTV and major airways (trachea and main/lobar bronchi)			V-19 Gy <37%	Pneumonitis
Liver minus GTV	Contour right and left lobes as 1 structure including all parenchymal liver tissue but excluding the GTV and major draining ducts, extrahepatic portal vein, and gall bladder	700*	32		Basic liver function
Renal cortex (right and left)	Contour right and left kidney as 1 structure, including all parenchymal capsular tissue but exluding the renal hilum/vascular trunk (see above)	200*	26		Basic renal function

* One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Abbreviations: CT = computed tomography; GTV = gross target volume; PTV = planning target volume; TM = temporomandibular.

Table 11 Timmerman tables, 30 fractions (conventional fractionation)—Timmerman, 8-2021

Serial tissue	Contouring instructions	Volume	Volume max (Gy)	Max point dose (Gy)	Endpoint (grade ≥3)
Optic pathway	One structure both sides from posterior globe, including chiasm, to proximal optic radiations	<0.5 cm^3	44	52	Neuritis
Eye (retina)	Each side separately, entire globe	Mean dose	<38	45	Retinitis
Lens	Each side separately			10	Cataract
Eyelid, meibomian glands (one side)	Each side separately, upper and lower lid as 1 structure			32	Dry eye syndrome
Lacrimal gland (one side)	Each side separately	<1 cm^3	20	36	Lack of tears
Cochlea	Each side separately, include at least 3 CT slices	<0.5 cm^3	36	40	Hearing loss
Brain stem (not medulla)	Superiorly from incisura, midbrain and pons only, 1 structure	<5 cm^3	52	60	Cranial neuropathy
Spinal cord and medulla	For medulla: starting at inferior pons to foramen magnum; for cord: entire bony canal including at least 10 cm superior and inferior to PTV	<5 cm^3	47.4	52.8	Myelitis
Salivary gland (one side)	Each parotid gland separately	<7 cm^3 Mean dose	20 <26	32	Xerostomia
Larynx	Starting 1 cm above first appearance of true vocal cord, include entire cord, arytenoid muscles, corniculate and arytenoid cartilages, and portions of thyroid cartilage abutting these structures, ending at the first appearance of the cricothyroid ligament	<3 cm^3	39	63	Necrosis/edema
TM joint	Each side separately starting at the superior articular surface near the zygoma bone and ending at the notch at the superior part of the ramus of the mandible	<1 cm^3	60	65	Inflammation
Cauda equina	Based on the bony limits of the spinal canal starting superiorly at the bottom of the spinal cord (typically around L2) and ending at the inferior extent of the thecal sac (typically S3)	<5 cm^3	50	60	Neuritis
Sacral plexus	Outlining the space defined medially by the sacral foramina from S1-S3, including contouring within the sacral foramina, posteriorly along the limits of the true pelvis, laterally to 2-3 cm lateral to the sacral foramina, and anteriorly about 3-5 mm from the posterior limits of the contour	<5 cm^3	50	60	Neuropathy
Esophagus	Include the mucosal, submucosa, and all muscular layers out to the fatty adventitia at least 10 cm superior and inferior to PTV	<5 cm^3	51	60	Esophagitis
Brachial plexus	Each side separately from the spinal nerves exiting the neuroforamina from around C5 to T2 to include only the major trunks of the brachial plexus, using the subclavian and axillary vessels as a surrogate for identifying its location, extending proximally at the bifurcation of the brachiocephalic trunk into the jugular/subclavian veins (or carotid/subclavian arteries) and following along the route of the subclavian vein to the axillary vein ending after the neurovascular structures cross the second rib	<3 cm^3	62	66	Neuropathy
Heart/pericardium	Contoured along with the pericardial sac; the superior aspect (or base) for purposes of contouring will begin at the level of the inferior aspect of the aortic arch (aorto-pulmonary window) and extend inferiorly to the apex of the heart	<15 cm^3 <20% of total heart volume	60 40	68	Pericarditis, heart problems
Great vessels	The wall and lumen of the named vessel at least 10 cm superior and inferior to PTV	<10 cm^3	60	76	Aneurysm
Trachea and large bronchus	Contour the trachea and cartilage rings starting 10 cm superior to the PTV, extending inferiorly to the bronchi, ending at the first bifurcation of the named lobar bronchus	<5 cm^3	60	69	Impairment of pulmonary toilet
Skin	The outer 0.5 cm of the body surface anywhere within the whole body contour	<10 cm^3	70	76	Ulceration
Stomach	The entire stomach wall and the gastric contents included from the GE junction to the proximal duodenum at the pyloris	<50 cm^3	45	60	Ulceration/fistula
Duodenum	The entire duodenal wall and lumen from the pyloris to the duodenojejunal flexure	<5 cm^3	45	60	Ulceration

(*Continued*)

Table 11 (*Continued*)

Serial tissue	Contouring instructions	Volume	Volume max (Gy)	Max point dose (Gy)	Endpoint (grade ≥3)
Jejunum/ileum	Any and all loops of small bowel as 1 structure within 10 cm of the PTV in any direction	<120 cm^3	45	54	Enteritis/ obstruction
Renal hilum/vascular trunk	Each side separately, including major calyces, renal pelvis, and proximal renal artery medially to the aorta	15 cm^3	42		Malignant hypertension
Colon	One structure, including wall and contents of lumen starting 10 cm superior to PTV and ending 10 cm below PTV	<20 cm^3	54	70	Colitis/fistula
Rectum (including stool)	One structure, including wall of rectum and all contents in lumen; start contouring 10 cm superior to PTV then inferior to anal sphincter	<10 cm^3 <20 cm^3 <30 cm^3 <40 cm^3	75 70 65 60	79	Proctitis/fistula
Bladder (with urine)	Contour the bladder wall and all urine, ending inferiorly at the base of the prostate	<90 cm^3 <150 cm^3	70 65	79	Cystitis/fistula
Bladder (suprapubic wall)	Contour the anterior inferior wall resting above and around the superior aspect of the pubic bone starting at the prostate inferiorly and extending 2-3 cm superiorly from there	<5 cm^3	30	60	Dysuria
Penile bulb	Contour starting superiorly at the inferior aspect of the pelvic diaphragm (urethral sphincter) and extending inferiorly and anteriorly up to 3 cm	<3 cm^3	48	56	Erectile dysfunction
Femoral heads	Contour both right and left separately	<10 cm^3	48	56	Necrosis
Growth plate (in pediatric patient)	Contour a single plate irradiated			4-6 (5% risk) 12 (100% risk)	Growth arrest
Parallel tissue		**Critical volume (cm^3)**	**Critical volume dose max (Gy)**	**Other constraints**	**Endpoint (grade ≥3)**
Lung (right and left) minus GTV	Contour right and left lung as 1 structure, including all parenchymal lung tissue but excluding the GTV and major airways (trachea and main/lobar bronchi)	1500 for males and 950 for females*	18		Basic lung function
Lung (right and left) minus GTV	Contour right and left lung as 1 structure, including all parenchymal lung tissue but excluding the GTV and major airways (trachea and main/lobar bronchi)			V-20 Gy <37%	Pneumonitis
Liver minus GTV	Contour right and left lobes as 1 structure, including all parenchymal liver tissue but excluding the GTV and major draining ducts, extrahepatic portal vein, and gall bladder	700*	36		Basic liver function
Renal cortex (right and left)	Contour right and left kidney as 1 structure, including all parenchymal capsular tissue but excluding the renal hilum/vascular trunk (see above)	200*	27		Basic renal function

* One-third of the "native" total organ volume (before any resection or volume reducing disease), whichever is greater.

Abbreviations: CT = computed tomography; GE = gastroesophageal; GTV = gross target volume; PTV = planning target volume; TM = temporomandibular.

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